

Temporal Context for Fluoroscopic Catheter and Guidewire Segmentation in Endovascular Robotics

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Abstract—Reliable catheter and guidewire localization in fluoroscopy could provide visual feedback for endovascular robotic systems. This study compared single-frame and temporal segmentation models on the CathAction human subset to evaluate the effects of training augmentation and prior-frame context. All models were evaluated on the same 580 test frames for which the required temporal inputs were available. U-Net achieved mean Dice scores of 0.623 without augmentation and 0.666 with augmentation. SegFormer-B0, stacked three-frame U-Net, and shared-encoder temporal U-Net achieved mean Dice scores of 0.622, 0.616, and 0.666, respectively. Neither temporal model exceeded the augmented single-frame U-Net on mean Dice, indicating that prior-frame context alone did not improve mean per-frame Dice in this experiment.

Index Terms—catheter and guidewire segmentation, endovascular robotics, fluoroscopy, medical image segmentation, temporal context

I. INTRODUCTION

Catheter and guidewire segmentation in fluoroscopy can provide an image-based estimate of device location, which could serve as a perception signal for endovascular robotic systems. The task is challenging because these devices are thin and can have low contrast, while segmentation models may also predict foreground outside the annotated device region.

Using the CathAction human subset [1], this study investigated two questions: whether training augmentation improves single-frame catheter and guidewire segmentation, and whether causal temporal context from earlier fluoroscopy frames improves segmentation beyond an augmented single-frame baseline. The evaluated architectures included U-Net [2], SegFormer-B0 [3], and two temporal U-Net variants. The selection of U-Net and SegFormer was informed by prior work on fluoroscopic catheter tracking [4]. The scope of this study was limited to image segmentation; clinical performance and robotic control were not evaluated.

II. METHODS

The CathAction human-subset images were divided into 3,501 training, 744 validation, and 604 test filenames across 46, 4, and 5 filename-derived sequences, respectively. A sequence was identified from the filename while ignoring the final frame index, allowing neighboring frames to remain in the same train, validation, or test split.

CathAction catheter and guidewire labels were combined into one binary foreground mask, referred to here as device segmentation.

TABLE I
MATCHED TEST SEGMENTATION RESULTS

Configuration	Dice	IoU	Prec.	Recall
U-Net, no aug.	0.623	0.492	0.654	0.697
U-Net, augmented	0.666	0.541	0.639	0.786
SegFormer-B0	0.622	0.481	0.570	0.786
Stacked 3-frame U-Net	0.616	0.486	0.628	0.748
Shared-encoder U-Net	0.666	0.541	0.646	0.775

Scores are means over the same 580 temporally eligible test images. Values are rounded to three decimals; bold indicates the highest rounded value for each metric, including ties.

The evaluated models included U-Net with and without augmentation, augmented SegFormer-B0, and two augmented temporal U-Net variants. Training augmentation included rotations up to $\pm 10^\circ$, horizontal flips, brightness/contrast and gamma changes, Gaussian noise and blur, and occasional rectangular occlusion. Rotations and flips were applied jointly to images and masks; temporal inputs received the same spatial transformation.

Both temporal models used grayscale frames at $t - 4$, $t - 2$, and t (about 167, 83, and 0 ms at 24 FPS) to predict the device mask at t , without future frames. The stacked temporal U-Net treated the three frames as separate input channels.

The shared-encoder temporal U-Net processed each frame using the same encoder. At each encoder resolution, it fused the ordered features from the three frames together with feature differences between the current frame and the two earlier frames.

All configurations were evaluated on the same 580 test frames from five filename-derived sequences. These were the frames among the 604 test filenames for which both $t - 4$ and $t - 2$ were available. Evaluation used 256×256 images and a probability threshold of 0.5.

Mean per-frame Dice was the primary endpoint, with intersection over union (IoU), precision, and recall as secondary metrics. Paired comparisons used 5,000 whole-sequence bootstrap resamples to account for correlation among frames from the same sequence.

III. RESULTS

The augmented U-Net achieved a mean Dice score of 0.666 compared with 0.623 without augmentation, an observed difference of 0.043 (95% sequence-bootstrap CI, -0.013 to

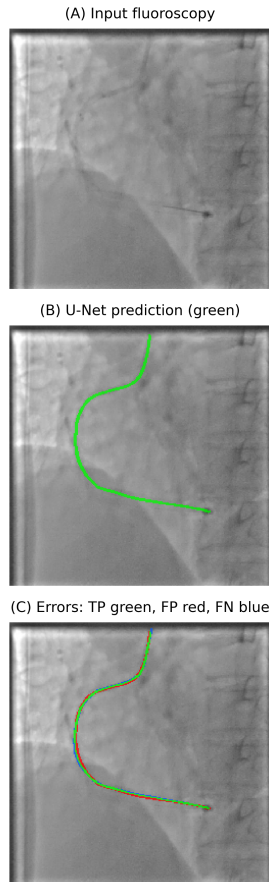


Fig. 1. Qualitative augmented U-Net example. (A) Input fluoroscopy. (B) Predicted catheter/guidewire mask overlay. (C) Comparison with the reference mask: green = TP, red = FP, and blue = FN. Example Dice = 0.710 at threshold 0.5; this value does not represent mean test performance.

0.073). Recall increased from 0.697 to 0.786, while precision decreased from 0.654 to 0.639.

SegFormer-B0 and stacked three-frame U-Net achieved mean Dice scores of 0.622 and 0.616, respectively. Relative to the augmented U-Net, the stacked temporal model had a mean Dice difference of -0.050 (95% sequence-bootstrap CI, -0.086 to 0.004). The shared-encoder temporal U-Net achieved 0.666, with a difference of -0.0001 (95% CI, -0.009 to 0.016). Thus, neither temporal model showed a mean Dice improvement over the augmented single-frame U-Net.

Fig. 1 shows an example prediction and its pixel-level agreement with the reference annotation.

IV. DISCUSSION AND CONCLUSION

In this experiment, the augmented U-Net achieved higher observed mean Dice and recall than the non-augmented U-Net. However, the sequence-bootstrap confidence interval for the Dice difference included zero, so this experiment does not provide strong evidence that augmentation consistently improves Dice.

The inclusion of earlier frames did not improve mean per-frame Dice in the evaluated temporal models. The stacked

temporal U-Net had lower observed mean Dice than the augmented single-frame model, although its 95% sequence-bootstrap confidence interval included zero. The shared-encoder temporal U-Net produced nearly the same mean Dice, with a paired difference close to zero and a confidence interval spanning zero. These results suggest that simply providing earlier frames was not enough to improve mean Dice in the models tested here.

Future work could test methods that explicitly account for motion between fluoroscopy frames or encourage predictions to remain consistent over time. These methods may be more useful when the catheter or guidewire is difficult to see in a single frame.

The study is also limited by five test sequences, one training run per configuration, and fewer eligible training examples for the temporal models. Repeated training runs and evaluation on additional independent recordings are needed to determine whether the observed differences are consistent.

For robotic use, segmentation accuracy alone is not enough. Future evaluation should also measure whether predictions remain stable over time, respond quickly when the catheter moves, and recover after missed detections. After further validation, device segmentation could be evaluated as visual feedback for an endovascular robotic platform [5] in vascular phantom experiments.

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